

8-26 ALTAM Additional practice

Question 28

The annual probability transition matrix for an insured age $60+k, k=0,1,\dots,9$ is:

$$\begin{array}{c} 0 \quad 1 \quad 2 \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{pmatrix} 0.90-0.01k & 0.05 & 0.05+0.01k \\ 0.70-0.01k & 0.20 & 0.10+0.01k \\ 0 & 0 & 1 \end{pmatrix} \end{array}$$

Calculate ${}_2p_{60}^{01}$.

Spring 2023 Exam ALTAM

(10 points) You are given the following information about a portfolio of in-force policies.

- (i) Policyholders are either Healthy (State 0), Temporarily Disabled (State 1), Permanently Disabled (State 2), or Dead (State 3).
- (ii) At time 0, there are 14 Healthy lives, 0 Temporarily Disabled lives, and 1 Permanently Disabled life.
- (iii) All 15 lives are independent.
- (iv) Transitions are modeled as a homogeneous Markov model.
- (v) The annual transition matrix is

$$P = \begin{pmatrix} 0.80 & 0.10 & 0.05 & 0.05 \\ 0.20 & 0.50 & 0.10 & 0.20 \\ 0 & 0 & 0.70 & 0.30 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- (vi) The two-year transition matrix, but with some missing values denoted by --, is

$${}_2P = \begin{pmatrix} 0.660 & -- & 0.085 & -- \\ 0.260 & 0.270 & 0.130 & -- \\ 0 & 0 & -- & -- \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- (a) (2 points) Calculate the missing values in the two-year transition matrix.